Systems Engineering Take-Home Exercise

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Timebox: 45 minutes maximum

Context

You have joined a startup building a safety-critical autonomous outdoor system. A customer demo is in 6 weeks. The team has strong mechanical, electrical, controls, software, and test engineers, but system-level requirements and verification are still loose.

Leadership has said:

The system needs to reliably detect and track small moving objects in cluttered outdoor conditions.

Setup must be simple enough that operators, not senior engineers, can run it.

The next demo needs to produce evidence we can use for a real go/no-go decision.

The system should be portable by two people and operate for several hours untethered.

Known issues:

Tracking is sometimes lost after vibration or brief occlusion.

Calibration currently takes about 25 minutes and usually needs a senior engineer.

Test logs are inconsistent; some runs are missing timestamps or metadata.

Ownership is unclear between software, controls, and test for reacquisition and data quality.

Deliverable

Send back a maximum of 2 pages or 5 slides. Rough is fine.

Please Cover

The 6 most important system requirements or success criteria you would define first.

For the top 3, how you would verify them: method, pass/fail criteria, owner, and evidence.

The 3 biggest gaps, risks, or assumptions you would surface immediately.

What you would do in your first week to drive clarity and closure.

2 things you would deliberately not formalize yet, and why.

Make reasonable assumptions and state them. No outside research is needed. We care more about judgment, prioritization, and clarity than completeness.

Requirements Table: Created from given scenario and stated user needs

Functional Area	Requirement	Notes/ Clarification	Owning Team	Validation/ Test
Tracking	The system shall be able to identify objects with minimum size 21x10x9cm up to distance of 1 mile from system	Based on quick search of drone size	Controls	Demonstration (Test Scenario 1)
Tracking	The system shall be able to track objects with minimum size 21x10x9cm with accuracy of +-1 degree positional error in both axes	Assuming system uses two sets of rotational coordinates, left/right and up/down	Controls	Demonstration (Test Scenario 1)
Tracking	The system shall be able to track objects with minimum size 21x10x9cm accelerating 10 m/s^2	Acceleration numbers are a good faith estimate of drone performance.	Controls	Demonstration (Test Scenario 1)
Tracking	If targeting tracking is lost, the system shall be able to re-acquire target object within 500 ms		Controls	Demonstration (Test Scenario 1)
Reliability	The system shall be resilient to mechanical vibration up to 30hz for 5 hours		Mechanical	Measurement (Test Scenario 2)
Reliability	The system shall be resilient to impulse up to 50 N*s for 5 hours	Based on quick search of average machine gun vibrational properties.	Mechanical	Measurement (Test Scenario 2)
Operational	The total system weight shall not exceed 400 lbs	Assuming low end male soldier deadlift of 200lbs. X2 for two operators	Mechanical	Measurement
Operational	Given a full state of charge, the system shall be able to operate at full functionality for 5 hours	“several” = roughly 5 hrs, specific time negotiable with stakeholders. Full functionality is defined as adhering to all requirements	Electrical	Demonstration
Operational	The system mechatronic software calibration routine shall be autonomous	“autonomous” requires additional elaboration, will need to discuss with client about specifics	Software/Controls	Demonstration
Diagnostic	The system shall include logs of performance and errors		Software	Observation
Diagnostic	The system logs shall contain timestamps		Software	Observation
Diagnostic	The system logs shall contain metadata		Software	Observation
Diagnostic	The system logs shall contain system state information at time of operation		Software	Observation
Diagnostic	The system logs shall contain sensor data		Software	Observation

Highlighted requirements are most important. These requirements define the core functionality of the system and are the highest value to our clients. The rest are supplementary and supportive of core functionality.

Testing Scenarios

Test Scenario 1:

- Part 1: Controls simulation environment
  - Gazebo or other physics environment, simulating system and targets.
  - Test drones of various sizes, distances, and accelerations
  - Include cover items/debris to purposely break tracking and trigger re-acquisition
  - Add simulated vibrations and/or sensor noise
  - Measurement objectives:
    - Tracking accuracy
    - Tracking consistency
    - Re-acquisition response time
- Part 2: Physical mechatronic test bench
  - Measuring real response time to commands
  - Measuring real tracking efficacy and accuracy
  - Looking for performance degradation under load or continuous usage.
  - Necessary because of things difficult to simulate, like friction in the system or heat from continuous firing

Test Scenario 2:

- Design review
  - Check mechanical structure natural frequencies and avoid anything that can resonate with the vibrational loads from the guns.
  - Review reliability and expected performance of off the shelf components.
- Physical vibration test bench
  - Physical test harness that we install the system into, mimicking real world conditions
  - Assess system performance under disturbance and load
  - Assess mechanical structure integrity under requirement loads
  - Extend test scenarios to failure, begin tracking MTBF for components and subsystems

Some test scenarios can cover multiple requirements. For example, an extended controls tracking test can also be an electronics endurance test. (Does it actually last 5 hours?)

Gaps, Risks, Assumptions

Gaps

- Create Interface Control Documents for system level requirements satisfied by multiple teams
- Still needs a more comprehensive testing/validation strategy
- Subsystem requirements decomposition still needed
- Safety requirements still needed

Risks

- Performance requirements may be too strict
- Requirements may not cover full use cases in real world situations
- Failure modes not addressed (highly important given this is a weapons system.)

Assumptions

- Specific requirement values were my best guess estimate. Further conversations with clients/stakeholders may refine the values.
- Knowledge of the skill level of the operator is still unknown, and what training is expected/given to operators before usage

First Week Alignment

- Centralize systems engineering documentation/models
- Connect requirements to engineering effort.
  - E.g. If the team uses jira for work tracking, linking all engineering effort to a system level requirement.
- Alignment meeting/call with specific engineering team leads from: mechanical, electrical, controls, software, test teams
  - Communicate requirement expectations and validation plans
  - Assess current adherence to requirements (which ones are met and which are not)
  - Address concerns/negotiate requirements
  - Discuss timeline feasibility (6 weeks)

Not to formalize yet

- Full calibration sequence requirements, this is dependent on expected interfacing with operators and other equipment and final formalized use cases. I expect the clients will also give further requirements regarding reliability performance under different temperatures/humidities/environments, etc. etc.
- Overly document and formalize engineering processes. Systems engineering is meant to bring clarity, not drag everyone down in excessive paperwork. So long as problems and objectives are appropriately defined for various teams, we can ensure efficient usage of engineering resources, and keep ourselves lean and agile.
- Further requirements decomposition of tracking functionality. There may be opportunities for controls to improve performance as part of the testing process. Would want to discuss with controls team.
    - i.e. an ML model that improves over time with more usage/testing, and so doesn't fit into a "clean" validation model initially.
- Defining and formalizing failure effect modes. Failure modes are solution dependent, and so some aspects of the system need to be decided before failure analysis can take place.
    - E.g. the selection of sensors on the system will determine what types of failures we should design for, and what level of redundancy is required.